Solutions to Homework 5

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Problem 1

Consider the class of Lipschitz functions on the interval $[0,1]$ defined as

$$\mathcal{F}_L \coloneqq \left\{ f : [0,1] \to \mathbb{R} \mid f(0) = 0 \text{ and } |f(x) - f(x')| \le L|x - x'| \quad \forall x, x' \in [0,1] \right\},$$

equipped with the metric associated with the sup-norm

$$d(f, f') \coloneqq \sup_{x \in [0,1]} |f(x) - f(x')|.$$

Prove that the metric entropy of the class $\mathcal{F}_L$ with respect to the sup-norm scales as

$$\log_2 \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) \asymp \frac{L}{\epsilon} \quad \text{for suitable small } \epsilon > 0,$$

where $\asymp$ denotes $\log_2 \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) = O(\frac{L}{\epsilon})$ and $\frac{L}{\epsilon} = O(\log_2 \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon))$.

Proof. A typical method to this problem is to find an ϵ-covering and a 2ϵ-packing of the class $\mathcal{F}_L$ consisting of the same number of elements K_ϵ. Then we have

$$\mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) \le K_\epsilon \quad \text{and} \quad \mathcal{P}(\mathcal{F}_L, ||\cdot||_\infty, 2\epsilon) \ge K_\epsilon,$$

and hence by applying the inequality

$$\mathcal{P}(\mathcal{F}_L, ||\cdot||_\infty, 2\epsilon) \le \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon),$$

we obtain

$$\mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) = \mathcal{P}(\mathcal{F}_L, ||\cdot||_\infty, 2\epsilon) = K_\epsilon.$$

ϵ-covering step: We need to construct an ϵ-covering of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$. To do so, defining $n \coloneqq \lfloor \frac{1}{\epsilon} \rfloor$, we divide the interval $[0,1]$ into $n+1$ segments $I_k = [x_{k-1}, x_k]$ for $k = 1, 2, \ldots, n+1$ with

$$x_k = k\epsilon, \quad \text{for } k = 0, 1, \ldots, n, \quad \text{and} \quad x_{n+1} = 1.$$

Moreover, we define the function $\Phi : \mathbb{R} \to \mathbb{R}$ via

$$\Phi(u) := \begin{cases} 0 & \text{for } u < 0, \\ u & \text{for } 0 \le u \le 1, \\ 1 & \text{otherwise.} \end{cases}$$

For each binary sequence $\beta \in \{-1, +1\}^n$, we may define a function $f_\beta : [0, 1] \to [-L, L]$ via

$$f_\beta(y) = \sum_{k=1}^{n} \beta_k L \epsilon \Phi\left(\frac{y - x_k}{\epsilon}\right).$$

By construction, each function is piecewise linear and continuous, with slope either $+L$ or $-L$ over each of the intervals I_k for $k = 2, \ldots, n+1$, and constant on the remaining interval I_1, see Figure 1 for an illustration.

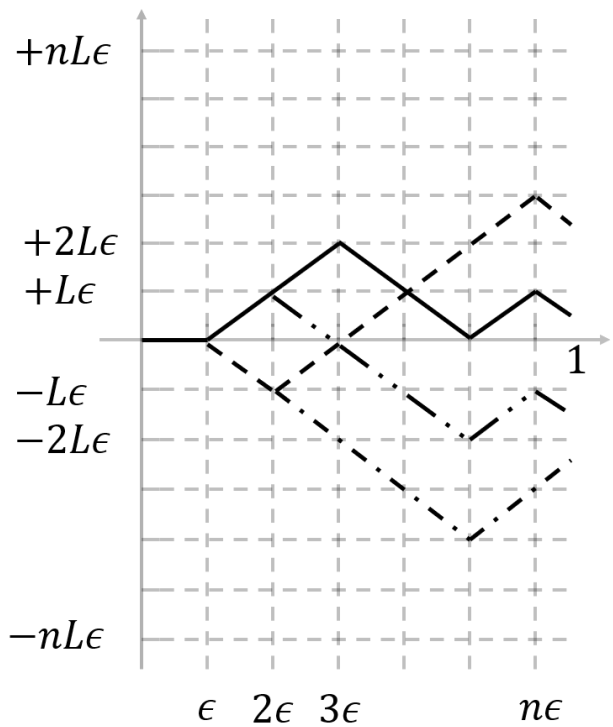

Figure 1: The function class $\{f_\beta \mid \beta \in \{-1, +1\}^n\}$ used to construct a covering of the class $\mathcal{F}_L$.

We first show that for any choice of β, $f_\beta \in \mathcal{F}_L$. It's obvious that $f_\beta(0) = 0$. Without loss of

generality, we let $y \in (x_p, x_{p+1}]$ and $y' \in (x_q, x_{q+1}]$ and $n \geq p > q \geq 0$. By definition of f_β, we have

$$\begin{aligned}
|f_\beta(y) - f_\beta(y')| &= \left| \sum_{k=1}^{p} \beta_k L\epsilon \Phi\left(\frac{y - x_k}{\epsilon}\right) - \sum_{k=1}^{q} \beta_k L\epsilon \Phi\left(\frac{y - x_k}{\epsilon}\right) \right| \\
&= L\epsilon \left| \sum_{k=q+1}^{p-1} \beta_k + \beta_p \left(\frac{y - x_p}{\epsilon}\right) + \beta_q \left(1 - \frac{y' - x_q}{\epsilon}\right) \right| \\
&\leq L\epsilon \left(\sum_{k=q+1}^{p-1} |\beta_k| + |\beta_p| \left(\frac{y - x_p}{\epsilon}\right) + |\beta_q| \left(1 - \frac{y' - x_q}{\epsilon}\right) \right) \\
&= L\epsilon \left(p - q + \left(\frac{y - x_p}{\epsilon}\right) - \left(\frac{y' - x_q}{\epsilon}\right) \right) \\
&= L\epsilon \left(p - q - \left(\frac{x_p - x_q}{\epsilon}\right) + \left(\frac{y - y'}{\epsilon}\right) \right) \\
&= L|y - y'|.
\end{aligned}$$

Hence for all β, $f_\beta \in \mathcal{F}_L$. Next we will prove by induction on $K = 1, \ldots, n+1$ that for every $f \in \mathcal{F}_L$ there is a β such that

$$\sup_{x \in \bigcup_{k=1}^{K} I_k} |f(x) - f_\beta(x)| \leq L\epsilon.$$

For $K = 1$, for any choice of β, we have $f_\beta(x) = 0$, $\forall x \in [0, \epsilon]$. By L-Lipschitz assumption, for every $f \in \mathcal{F}_L$, we have

$$|f(x) - f_\beta(x)| = |f(x)| = |f(x) - f(0)| \leq L|x - 0| \leq L\epsilon,$$

which proves the property for $K = 1$.

Assuming the property is satisfied up to some rank K, we consider the rank $K + 1$. Given a function $f \in \mathcal{F}_L$, by induction hypothesis, let β be such that

$$\sup_{x \in \bigcup_{k=1}^{K} I_k} |f(x) - f_\beta(x)| \leq L\epsilon, \ \forall x \in [0, x_K].$$

By L-Lipschitz assumption, $|f(x_{K+1}) - f(x_K)| \leq L|x_{K+1} - x_K| = L\epsilon$. Then we have

$$f_\beta(x_K) - 2L\epsilon \leq f(x_K) - L\epsilon \leq f(x_{K+1}) \leq f(x_K) + L\epsilon \leq f_\beta(x_K) + 2L\epsilon.$$

Thus, either $f(x_{K+1}) \in [f_\beta(x_K) - 2L\epsilon, f_\beta(x_K)]$ or $f(x_{K+1}) \in [f_\beta(x_K), f_\beta(x_K) + 2L\epsilon]$. We treat only the first case, the method being the same for the second one. We take β such that the K-th term is -1(also note that we take β such that the K-th term is 1 for the second case). Assume that we have some $x \in [x_K, x_{K+1}]$ such that $f(x) < f_\beta(x) - L\epsilon$, then

$$\begin{aligned}
f(x_K) - f(x) &> (f_\beta(x_K) - L\epsilon) - (f_\beta(x) - L\epsilon) \\
&= f_\beta(x_K) - f_\beta(x) \\
&= L(x - x_K),
\end{aligned}$$

which is a contradiction with the L-Lipschitz assumption on f. On the contrary, if there is some $x \in [x_K, x_{K+1}]$ such that $f(x) > f_\beta(x) + L\epsilon$, using $f(x_{K+1}) \leq f_\beta(x_K)$, then

$$\begin{aligned} f(x) - f(x_{K+1}) &> (f_\beta(x) + L\epsilon) - f_\beta(x_K) \\ &= (f_\beta(x) + L\epsilon) - (f_\beta(x_{K+1}) + L\epsilon) \\ &= f_\beta(x) - f_\beta(x_{K+1}) \\ &= L(x_{K+1} - x), \end{aligned}$$

which is a contradiction with the L-Lipschitz assumption on f. Thus we have proven by induction that for all $K = 1, \ldots, n+1$, there is a β such that such that f_β is $L\epsilon$-close to a given f on $[0, x_K]$. In particular, set $K = n+1$ and we conclude that the set of functions $\{f_\beta \mid \beta \in \{-1,+1\}^n\}$ is an $L\epsilon$-covering of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$. Substituting $\frac{\epsilon}{L}$ for ϵ, then we get an ϵ-covering of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$.

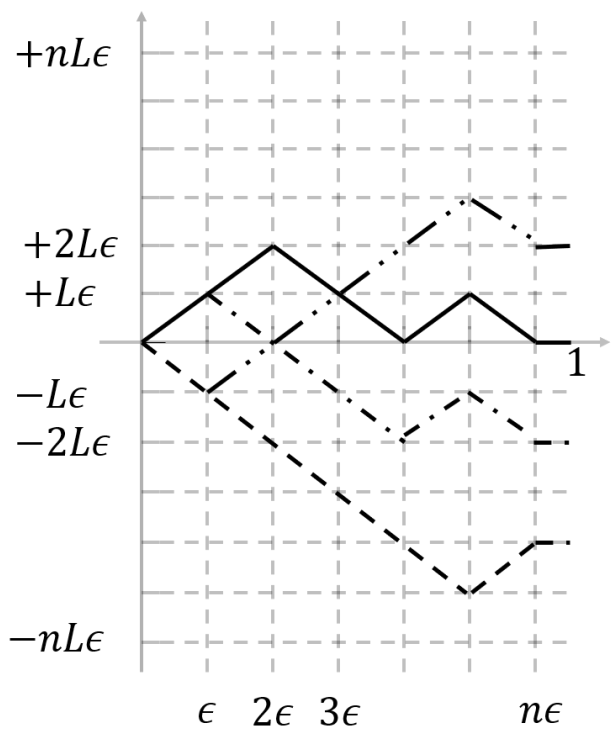

Figure 2: The function class $\{h_\beta \mid \beta \in \{-1,+1\}^n\}$ used to construct a packing of the class $\mathcal{F}_L$.

2ϵ-packing step: We need to construct a 2ϵ-packing of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$. To do so, for each binary sequence $\beta \in \{-1,+1\}^n$, we may define a function $h_\beta : [0,1] \to [-L, L]$ via

$$h_\beta(y) = \sum_{k=1}^{n} \beta_k L\epsilon\Phi\left(\frac{y - x_{k-1}}{\epsilon}\right).$$

By construction, each function is piecewise linear and continuous, with slope either $+L$ or $-L$ over each of the intervals I_k for $k = 1, \ldots, n$, and constant on the remaining interval I_{n+1}, see Figure 2 for an illustration.

By using the same method as in the ϵ-covering step, we can verify that $h_\beta(0) = 0$ and that $h_\beta \in \mathcal{F}_L$. Given a pair of distinct strings $\beta_1 \neq \beta_2$ and the two associated functions h_{β_1} and h_{β_2}, there is at least one interval I_k, with $1 \leq k \leq n$, where the functions start at the same point, and have opposite slopes over I_k. Since the functions have slope $+L$ and $-L$ over I_k, respectively, we are guaranteed that $||h_{\beta_1} - h_{\beta_2}||_\infty \geq 2L\epsilon$, showing that the set $\{h_\beta \mid \beta \in \{-1, +1\}^n\}$ forms a $2L\epsilon$-packing of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$. Substituting $\frac{\epsilon}{L}$ for ϵ, then we get a 2ϵ-packing of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$.

We have constructed an ϵ-covering and a 2ϵ-packing of the metric space $(\mathcal{F}_L, ||\cdot||_\infty)$, both of cardinality $K_\epsilon = 2^n = 2^{\lfloor \frac{L}{\epsilon} \rfloor}$. Therefore, using the following inequality

$$K_\epsilon \leq \mathcal{P}(\mathcal{F}_L, ||\cdot||_\infty, 2\epsilon) \leq \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) \leq K_\epsilon,$$

then we obtain

$$\log_2 \mathcal{N}(\mathcal{F}_L, ||\cdot||_\infty, \epsilon) \asymp \frac{L}{\epsilon}.$$

□